

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 13-100-3
CABLE TERMINATION FOR XLPE INSULATED CABLES 12/20 (24) kV (INDOOR/OUTDOOR)	6-05-2025

EDMS 13-100-3

SPECIFICATION

FOR

CABLE TERMINATIONS FOR XLPE INSULATED CABLES 12/20 (24) KV (INDOR/OUTDOOR)

Issue: May, 2025/ Rev- 3

- This revision contains option items that should be selected by EDC before bidding.

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1- SCOPE

This specification covers the minimum technical requirements for design, engineering, manufacture, testing, inspection, installation and performance of all kinds of terminations for medium voltage power cables, XLPE insulation, with the following different constructions:

- Three core and single core
 - Indoor and outdoor
 - Copper and aluminum conductors
- The termination for the level of 11Kv system (12/20 Kv cable) should meet the requirements of the 24Kv Level.
- The termination kits should be supplied completely with all relevant components
- Unless otherwise specified termination should comply with all relevant standards described in table (1)
- This specification does not cover Accessories for special applications, such as aerial cables, submarine or ship cables or hazardous situations (explosive environments, fire-resistant cables or seismic conditions).

2- DEFINITIONS

2.1. Indoor Termination

A device fitted to the end of a cable to ensure electrical connection with other parts of the system and to maintain the insulation up to the point of connection, where termination intended for use where it is not exposed to either solar radiation or weathering.

2.2. Outdoor Termination

A device fitted to the end of a cable to ensure electrical connection with other parts of the system and to maintain the insulation up to the point of connection, where termination intended for use where it is exposed to either solar radiation or weathering.

2.3. Conductor Temperature:

The termination should be suitable for use on cable having maximum conductor temperature of 90°C at normal conditions, and maximum temperature of the cable conductors under short-circuit condition of 250°C.

2.4. Rated voltage: U_0/U (U_m)

The rated voltages $U_0/U(U_m)$ of the cable termination considered in this specification is 12/20 (24) Kv.

- U_0 : is the rated power frequency voltage between conductor and earth or metallic screen for which the cable is designed (phase voltage).
- U : is the rated power frequency voltage between conductors for which the cable is designed (line voltage);
- U_m : is the maximum value of the "highest system voltage" for which the equipment may be used.

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3- APPLICABLE STANDARDS

The Terminations covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification should be clearly set out in the offer.

S. No.	Cross Reference	Description
1.	EDMS 15-300	COMPRESSION TYPE LUGS FOR CABLES UP TO 24 kV

Table (1)

S. No.	Standard No.	Description
2.	IEC 60502-2	Cables for rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
3.	IEC 60502-4	Test requirements on accessories for cables with rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
4.	IEC 61442	Test methods for accessories for power cables with rated voltages from 6 kV ($U_m = 7,2$ kV) up to 30 kV ($U_m = 36$ kV)
5.	CENELEC HD 629.1S3	Test requirements on accessories for use on power cables rated voltage from 3.6/6(7.2) kV up to 20.8/36(42) Kv - Part 1
6.	IEC 60228	Conductors of insulated cables
7.	IEC 60060	High-voltage test techniques
8.	IEC 60230	Impulse tests on cables and their accessories
9.	IEC 60885-3	Electrical test methods for electric cables Part 3: Test methods for partial discharge measurements on lengths of extruded power cables
10.	VDE 0278-629-1	Test requirements on accessories for use on power cables of rated voltage from 3,6/6 (7,2) Kv up to 20,8/36 (42) kV Part 1: accessories for extruded insulation
11.	IEEE 048	Standards for test procedure and requirements for alternating-current cable Termination used on shielded cables.

4- ENVIRONMENTAL CONDITIONS

The performance of the Termination components in the kits should be guaranteed for following the environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer.

Ambient temperature	-5°C to +45°C (50°C is an option according to --EDC requirements),
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level

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5- SERVICE CONDITIONS

- The sunlight is very intense and the temperature of exposed 70°C.
- The maximum solar radiation intensity on a horizontal surface is 1000 w/m².
- Severe surface contamination from salt, dust, heavy industrial pollution, sandstorms, fine dust often carried in suspension in the atmosphere even on calm days, can be expected.

6- SYSTEM DETAILS :

Frequency	50 HZ
Number of phases	1 or 3
Nominal voltage	11 KV
Highest voltage	12 KV
Maximum prospective Fault level	25 KA for Duration of 1 sec.

7- TERMINATION KIT MATERIAL, RATINGS, AND PERFORMANCE

- 7.1. The Termination kits should be supplied completely with all relevant components including the main insulation part(s), DIN or equivalent standards Lugs (tinned copper), Braid of sufficient length, cleaning solvent, solder, outer jacket tubes , and/or any special accessories required.
- 7.2. The material for bonding the cable screen, the cable armouring with the termination screen (copper braid or mesh) should be suitable for fault levels specified in clause 6 .
- 7.3. The cross-sectional area of the copper mesh or braid should be at least equal to the effective cross-sectional area of the cable copper screen with a minimum size as indicated in the following tables.

Minimum Copper Screen of termination for Single & three-Phase Cable 12/20 Kv

C.S.A (mm ²)	35	50	70	95	120	150	185	240	300	400	500	630	800	1000
Total Copper weight per termination (gm/m)	130					200				315				
Min. length of the copper Baird per phase cm	90													

7.4. Each kit should be supplied with installation instructions in Arabic (and English).

7.5. Termination should in general be compatible with cables used by ...EDC.

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7.6. Testing of the termination should be meet the requirements described in the relevant clause and either requirements of IEC, or VDE tests or according to CENELEC.

7.7. The minimum guaranteed performance should be suitable with the environmental conditions mentioned in item (4).

7.8. The 24 KV termination should be suitable for use with the following type of cables: 12/20 KV, one/three core, under-ground power cables, Aluminum/ Copper stranded-compacted circular conductor, extruded semi-conducting layer , XLPE insulation, extruded semi-conducting layer well bonded, copper tape screening, filler, separation sheath ST2 (bedding) , double steel tape armour, black PVC over sheath, according to the requirements of IEC 60502-2.

7.9. The materials of the termination and the installation instructions should be compatible with the cable materials 12/20 (24) KV, hence the cable insulation , and semi-conductive screening layers are extruded type.

7.10. Lugs should be compression type (crimped).

7.11. The termination lugs for aluminum cables should be Bi-metallic lugs , and it can be one of the following types of BI metallic lugs :(as required by ...EDC)

- BI metallic Insert lugs (AL lug with copper ring)
- BI metallic Friction welded lugs (AL lug with copper palm)
- BI metallic tinned aluminum Alloy

7.12. The termination lugs for copper cables should be copper lugs.

7.13. ADDITIONAL ACCESORIESFOR TERMINATION KIT

1. Skirts must be coated inside (with protective release paper) with mastic or the terminations should be guaranteed (by the performance bond 10% of termination price) for 3 year from delivery to stores. (for outdoor heat-shrinkable termination) .

Note:

- Minimum Creepage distance for Indoor termination 2.5 cm/kv. (Reference voltage 12kv)
 - Minimum Creepage Distance for Outdoor termination 4.5 cm/kv (Reference voltage 12kv).
 - For 3core outdoor termination only supplier shall supply shrinkable tube each or two tubes per phase with total length not less than 40 cm with termination (for heat shrinkable termination supplier can be supply one tube per phase its total length plus skirts equal to creepage distance + 40 cm)
- 1- Cleaning solvent >100 c.c(m.l) + cleaning tissues (3 tissues per phase), or cleaning wipes as the following:
 - 12 wipes for 3 core / 6 for single core termination.
 - Wipe: fiber (20*20cm)
 - Solvent type: isopropyl or tetrachloroethylene
 2. Tin wire solder (with flux): length > 1m for 1 or 3 core cables, or any other applicable solution.
 3. Silicon grease: (6 Packets) each one ≥ 5 gm or (3 Packets)each one ≥ 10 gm (for 3 core cables).

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4. Galvanized steel clamps: 1 pcs for binding steel armour layer. (only applicable For Three Core cables).
5. Stainless clamps: 1 pcs for binding AL armour layer. (only for single Core cables).
6. Mastick paste (or tape) to be used as sealant at least at:
 - Between breakout & cable outer jacket.
 - Between non-tracking tubes and breakout necks. (for heat-shrinkable termination)
 - Underneath and over cu. braid, cable steel tape and cu. tape to cover welding point.
 - Shrinkable non-tracking tubes shall be marked with diam. before and after free shrink (for heat-shrinkable termination).
7. Red mastic to fill over lug barrel and end of cable insulation and gap between them (to seal ends of non-tracking tubes).
8. 3 pcs of sandpaper size 180 with length 40 cm, cloth type (non-easy cut).(for 3 cores)
9. 1 pcs of sandpaper size 180 with length 40 cm, cloth type (non-easy cut).(for 1 core)
10. 1 piece of sandpaper size 320 with length 40 cm, cloth type (non-easy cut).

8- MARKING AND IDENTIFICATION

All components of the termination kit should be distinctly and clearly marked with the manufacturer's name and reference numbers. The marking should be done before packing the component. Electrically conducting components should be marked 'conducting' clearly and permanently unless they are very small to accommodate respective marking.

All components should be capable of being stored without deterioration within the temperature range of -5°C to +45°C (+50 °C as option). Components or materials, if subjected to a shelf-life limitation, should have the final date of use prominently and permanently shown on all packages.

The electrical relevant components of cable accessories must be marked with the following data:

1. Manufacturer name
2. Production indicator for traceability
3. Cross-section, diameter, or application range should be marked

Packaging as well for each cable accessory should be printed with the following information:

1. Manufacturer's name
2. Manufacturer's Type No. / catalog number
3. Purchase order number/tender
4. Cable size/diameter
5. Application (indoor or outdoor)
6. Nominal Voltage (12/20Kv) / Max. Voltage (24Kv)
7. Year of manufacture
8. Date of expiry
9. Manufacturer's batch No. or any code for tracing the quality assurance records maintained by the manufacturer.
10. Shipping Mark Guide.

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11. Kit components for electrical stress control, and non-tracking tubes shall durably marked for identification.

12. Kit components for indoor/outdoor pre-moulded termination stress con modules should be also durably marked for identification

* It should be possible to read all information directly

9- PACKING

9.1. Packing should be robust and able to withstand very rough handling during transportation. It should be protected against ingress of moisture.

9.2. The details of method of packing should be submitted with the offer.

9.3. Each kit should be supplied with a packing note stating the quantities and descriptions of components contained in the kits, and the documents described below.

9.4. The kits should include:

- Termination Instructions of installation (Arabic and English) for the offered accessories which must include the steps of installation, the number of dies, detailed drawing of the lugs explaining the number & sequence of crimping and the width of each crimping, (Each crimping should be hexagonal).
- Instructions of installation (Arabic and English preferred) of the offered accessories.

10- INSPECTION AND TESTING

10.1. ROUTINE TESTS

10.1.1 The routine tests should be done as per latest revision of applicable standards mentioned in table No.1

10.1.2 Tests should be done by the manufacturer on each finished product or component of accessories in the manufacturer facility to check that each product/ component meets the specified requirements.

10.1.3 Routine tests for the cable termination should be like those of the cables:

- AC dry withstand voltage test
(54Kv for 1 min. as per IEEE 404 or 42 Kv for 5 min. as per IEC 60502-2)
- Partial Discharge test (the PD at 20 Kv should less than 5 pc)

10.1.4 vendor should keep records for the routine tests

10.1.5 vendor should keep records for the calibration of the instruments and testing facilities which using in the routine tests

10.2. SAMPLE AND INSPECTION TEST

10.2.1 Sample tests should be made by the manufacturer on samples of finished product or components taken from a finished product store, at a specified frequency, so as to verify that the finished product meets the specified requirements.

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- 10.2.2** Inspection test should be done in the manufacturer/vendor facilities in attendance ofEDC representatives per each order before delivery..
- 10.2.3** Inspection tests should be conducted as per submitted/approved inspection plan during the tender stage,
- 10.2.4** the inspection plan should include a visual examination on each component, there should be no visual defects in the material, manufacture, workmanship and markings.
- 10.2.5** The Tenderer should provide free of charge all the testing facilities required at work.
- 10.2.6** The routine tests should be done as per applicable standards up on manufacturer`s type tests standard.
- 10.2.7** The inspection sample should be 1% of the order quantity and should not be more than 5 pcs of each termination size.

10.3. ELECTRICAL TESTS AFTER INSTALLATION

tests made to demonstrate the integrity of the cable and its accessories as installed as per IEC 60502-2 clause 20.3.1 (AC testing):

- 10.3.1** By agreement between the purchaser and the contractor, an a.c. voltage test in accordance with IEC 60060-3 and in accordance with item a), b) or c) as below may be used:
- a)** Test for 15 min with the phase-to-phase voltage U , at a frequency between 20 Hz to 300 Hz shall be applied between the conductor and the metal screen/sheath;
 - b)** Test for 24 h with the normal rated voltage U of the system.
 - c)** Test for 15 min with the RMS rated voltage value of $3 U_0$ at a frequency of 0,1Hz applied between the conductor and the metal screen/sheath.

NOTE 1:

During the a.c. test, $\tan \delta$ and/or partial discharge may be monitored. 0

NOTE 2 :

For installations which have been in use, lower voltages and/or shorter durations may be used. Values should be negotiated, taking into account the age, environment, history of breakdowns and the purpose of carrying out the test.

10.4. TYPE TESTS:

- 10.4.1** Type Testing should be made to demonstrate satisfactory performance characteristics according to the latest standard to meet the intended application.
- 10.4.2** It is not necessary to repeat these tests, once successfully completed, unless changes are made in the materials, design or manufacturing process which might affect the performance characteristics
- 10.4.3** Cables used for testing shall comply with IEC 60502-2 and shall be of the same rated voltage as the accessories to be tested
- 10.4.4** Accessories should be assembled in the manner specified by the manufacturer's instructions.

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- 10.4.5 Type tests should be done in certified independent and approved test laboratory
- 10.4.6 Type test for cable termination should be conducted as per IEC60502-4 table 9 or CENELEC HD 629-1S3 standards.
- 10.4.7 Test method shall be as per IEC 61442 or CENELEC standards.
- 10.4.8 The test values and requirements should be as per IEC60502-4 table 17.
- 10.4.9 Type testing complying with the full requirements of CENELEC HD 629.1S3 is or VDE HD 629-1 both accepted too.
- 10.4.10 Type test certificates should be submitted with the offers (if not before submitted) to be reviewed by(...EDC), ...EDC should require its representative to assess the quality control and tests.
- 10.4.11 The test program dates should be advised by the contractor in advance to permit witnessing by ...EDC.
- 10.4.12 ...EDC reserves its right to send inspector for that purpose, and ...EDC representative should sign the test certificates in case of acceptance.
- 10.4.13 Tests described above should not release contractor from his full responsibility towards supplies, guarantee, and compliance with this specification.
- 10.4.14 Range of approval for the type test should be as per IEC 60502-4 clause 8.

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Table 9 – Test sequences and requirements for terminations

Tests ^a	Requirements	Test methods of IEC 61442:2023	Test sequences (see Figure 1)				
			1.1	1.2	1.3	1.4	1.5
1 AC voltage dry	5 min at 4,5 U_0 . No breakdown nor flashover.	Clause 5	X	X	X		
AC (wet)	1 min at 4 U_0 ^b . No breakdown nor flashover.	Clause 5	X				
2 Partial discharge at ambient	10 pC max. at 1,73 U_0 . ^c	Clause 8	X				
3 Impulse at θ_t	10 impulses of each polarity. No breakdown nor flashover.	Clause 7	X				
4 Heating cycles in air	Min. 60 cycles at θ_t and 2,5 U_0 .	10.2	X				
5 Immersion test ^b	10 cycles at θ_t .	10.4	X				
6 Partial discharge at θ_t and ambient temperature	10 pC max. at 1,73 U_0 . ^c	Clause 8	X				
7 Thermal short-circuit (screen) ^d	Two short-circuits at I_{sc} of the cable screen. No visible deterioration.	Clause 11		X	X		
8 Thermal short-circuit (conductor)	Two short-circuits to raise conductor to θ_{sc} of the cable. No visible deterioration.	Clause 12		X	X ^e		
9 Dynamic short-circuit ^f	One short-circuit at I_d . No visible deterioration.	Clause 13			X ^e		
10 Impulse at ambient	10 impulses of each polarity. No breakdown nor flashover.	Clause 7	X	X	X		
11 AC voltage dry	5 min at 4,5 U_0 . No breakdown nor flashover.	Clause 5	X	X	X		
12 Partial discharge at ambient	10 pC max. at 1,73 U_0 . ^c	Clause 8	X				
13 Humidity ^{g, h}	300 h at 1,25 U_0 , see Table 16.	Clause 14				X	
14 Salt fog ^{b, h}	1 000 h at 1,25 U_0 , see Table 16.	Clause 14					X
15 Examination according to Annex C	Shall be documented in the report.	–	X	X	X	X	X

^a The tests shall be carried out at ambient temperature, unless otherwise specified.

^b For outdoor terminations only.

^d This test is only required for terminations that are equipped with a connection to, or adaptor for, the metallic screen of the cable.

^e The thermal short-circuit may be combined with the dynamic short-circuit.

^f Applies to three-core cables only. For the value of the current I_d , see 9.2. Due consideration shall be given to ensure that the installation arrangement is representative for the application. The value of the class shall be declared by the manufacturer, in accordance with 9.2.

^g For indoor terminations only. Not required for compound-filled terminal boxes. Shrouded terminations shall be tested in a three-phase condition.

^h Not required for terminations having porcelain insulators.

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10.4.15 Extension of approval to a range wider than that given above should be achieved by carrying out the additional tests specified at IEC 60502-4 listed in Table 6 for smaller, and larger conductor cross sectional areas of the required range.

Table 6 - Extension of compliance and qualification scheme for non-range-tracking termination

Accessory size tested	Test sequences	Samples	Range of cross-sections qualified
Size < 95 mm ²	Table 14	2 (1- core) 1 (3- core)	Extension down to and including the size tested
Size > 300 mm ²	1.1 ^a	2 (1- core) 1 (3- core)	Extension up to and including the size tested

^a Test sequences refer to Table 9

Table 14 – Additional tests for smaller conductor cross-sectional areas

Tests ^a		Requirements	Test methods of IEC 61442:2023	Test sequences (see Figure 1, Figure 2, Figure 3)		
				1.1 ^b	2.1 ^c	3.1 ^d
1	AC voltage dry	5 min at 4,5 U ₀ . No breakdown nor flashover.	Clause 5	X	X	X
2	Partial discharge at ambient	10 pC max. at 1,73 U ₀ ^e	Clause 8	X	X	X
3	Impulse	10 impulses of each polarity. No breakdown nor flashover.	Clause 7	X	X	X
4	Heating cycles	No breakdown nor flashover. Terminations. Min. 10 cycles in air at θ_t^g and 2,5 U ₀ .	10.2 and 10.3	X	X	X
5	AC voltage dry	5 min at 4,5 U ₀ . No breakdown nor flashover.	Clause 5	X	X	X
6	Partial discharge at θ_t^g , and ambient temperature	10 pC max. at 1,73 U ₀ ^e .	Clause 8	X	X	X
7	Examination according to Annex C	Shall be documented in the report.	–	X	X	X

^a The tests shall be carried out at ambient temperature, unless otherwise specified.

^b Terminations: test half the number of samples given in Figure 1.

^g θ_t is the maximum cable conductor temperature in normal operation +5 K to 10 K.

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Table 17 – Summary of test voltages (see Clause 9)

Test	Test voltage	Rated voltage $U_0/U(U_m)$ kV				
		3,6/6 (7,2)	6/10 (12)	8,7/15 (17,5)	12/20 (24)	18/30 (36)
Humidity and salt fog	1,25 U_0	4,5	7,5	11	15	22,5
Partial discharge	1,73 U_0	6	10	15	20	30
Heating cycle voltage and AC voltage/500 h	2,5 U_0	9	15	22	30	45
AC voltage/1 min	4 U_0	14,5	24	35	48	72
AC voltage/5 min	4,5 U_0	16	27	39	54	81
Impulse (peak)	–	60	75	95	125	170

11- TENDER DOCUMENTS AND SUBMITTAL

The tenderer should submit with his offer a duly completed and signed package of all the required documents as:

- 11.1 A reference list showing that the proposed technique has been extensively used by recognized utility companies outside the manufacturer's country for 3 years (for foreign manufacturers only).
- 11.2 Copies of recently (latest) conducted type test certificates (electrical and mechanical) or qualification reports from certified independent and approved test laboratory for the offered terminations that verify all the requirements according to IEC 60502-4 or VDE 0278-629-1 or CENELEC HD 629-1S3 standards.
- 11.3 Data sheet for the lugs used within the accessories should include at least the following data:
 - 11.3.1 Lug material details (alloy, conductivity, chemical composition ...)
 - 11.3.2 Assembly technique,
 - 11.3.3 Tooling, dies and necessary setting,
 - 11.3.4 Preparation of contact surfaces, if applicable,
 - 11.3.5 Type, reference number and any other identification of the lug,
 - 11.3.6 Lug details of the type test approval, the standard with which they comply (i.e. IEC 61238-1) or DIN
- 11.4 The Factory Acceptance Test (FAT), which carried out at manufacturer/vendor facilities on each finished product and components of accessories to check that each product and components meets the specified requirements
- 11.5 A detailed list of inspection tests that have been carried out at manufacturer's facilities in attendance of the ...EDC representative(s).
- 11.6 The name of the laboratory where tests can be conducted
- 11.7 Complete technical details with dimensional drawings, showing description of material, the following data at least should be included:

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- 11.7.1. Name of manufacturer and manufacturing location.
- 11.7.2. Type, designation, or code.
- 11.7.3. Rated voltage,
- 11.7.4. Minimum and maximum cable insulation diameters,
- 11.7.5. Minimum and maximum cable cross-sections for this rated voltage
- 11.7.6. Applicable cable construction including the material and shape of cable conductor
- 11.7.7. Installation instructions (reference)
- 11.8 Catalogue and Instructions of installation (Arabic and English preferred) of the offered accessories. The catalogues must include the number of dies, detailed drawing of the lug explaining the number of crimping and the width of each crimping, each crimping should be hexagonal.
- 11.9 Details of storage, shelf life under Egypt service conditions mentioned in clause 6 (It must not be >30 months).
- 11.10 Lifetime of termination should not be less than 30 years starting from the date of operation.
- 11.11 One sample of complete kit will be submitted with the offer.
- 11.12 Details of any special tool requirement.
- 11.13 Schedule of guarantee duly completed and signed by the manufacturer and vendor/Agent, otherwise the tenderer should be rejected.
- 11.14 Kit content lists stating the quantities and descriptions of components, showing dimensions of lugs galvanized steel wrap, and tinned copper mesh or Braid.

12- INSTALLATION INSTRUCTIONS

- 12.1 The installation instructions should be clearly assigned to the respective termination according to the application of this Technical Specification.
- 12.2 All necessary steps for installing the termination are to be described in an easily readable manner including images for each step.
- 12.3 The installation instructions must be approved by ...EDC prior to tender.
- 12.4 The installation instructions must contain the required designations and sheath-opening (cutback) dimensions.
- 12.5 The termination components, the kit content list, and the respective installation instructions must be clearly and unambiguously identifiable.
- 12.6 The installation instructions should be enclosed in English and Arabic Language (where applicable) and are printed in color.
- 12.7 Any modifications to the assembly or the material, even insignificant, which have an effect on the installation instructions, must be clearly noted on each installation instruction and packaging unit for at least half a year (e.g. with a colored sticker with exact reference to which point was changed), as agreed with ...EDC.
- 12.8 The revision level, the date of creation or revision must be recognizable on the installation instructions.
- 12.9 The final pages of each installation instructions is a full-page completely assembled cable termination in a cut-away/sectional view showing all active components of the kits.

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13- TRAINING.

The Vendor should provide a sufficient practical training wherever applicable for the installation and commissioning of the terminations. (if required by ...EDC).

14- GUARANTEE:

The vendor shall guarantee supplies against all defects arising out of faulty design or workman ship or defective material for a period of two year from date of delivery to ... EDC stores, inspection and technical acceptance.

The vendor shall guarantee the uniformity of the products delivered with the approved samples.

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GUARANTEE TABLE FOR TERMINATION PRE-MOUIDED ,... X mm² , 24 KV

[illegible]

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11	Weight of silicon grease -----gm.	-----
12	Binding tapes : A-Two PVC tapes:- Length / thickness---- m./mm	----- -----
13	Mastic pastes : - A – 1 st type: type / number/ weight. each B – 2 nd type: type / number/ weight. each	----- ----- -----
14	Any other tape : type , length -----	
15	Attach details on rated short circuit current capacity : - a) Copper mesh or BRAID b) Armour c) Bonded mesh or BRAID with armour	
16	Copy of works routine test reports and if requested byEDC also batch / type test reports on the kits being supplied shall be cleared byEDC prior to shipment confirm compliance.	Yes / No
17	Confirm that visual inspection and routine tests shall be carried out on each kit -----	Yes / No
18	Confirm compliance to the appropriate requirements of table 2-----	Yes / No
19	Marking and packing:- a) Marking shall be to clause 4.4 ----- b) Packing shall be to clause 4.5 ----- c) Each kit with components shall be complete in itself and separately packed ----- d) Submit detailed method of packing	Yes / No Yes / No Yes / No Submitted / not subm.
20	Guarantee confirm compliance to clause 6.3 ----	Yes / No
21	For direct technical clarification from the manufacturer submit the details mentioned below of the persons responsible : - Name :----- Designation : ----- Address : ----- Telephone : ----- FAX : -----	

We guarantee above mentioned data for our offered material forEDC

Vendor / Agent

Important: Each tenderer shall submit with his offer the above tender " specifications for cableterminations ...x mm² 24 KV. With each page signed by the tenderer.

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This signature means that tenderer confirms verification of his offer for these specifications.
Tenders not accompanied with signed specifications shall not be considered.

SCHEDULE GUARANTEE

Schedule of deviations from -----Technical specification

This schedule be filled and signed by every tenderer. If there are no deviations or reservations.

This should be explicitly stated in this schedule.

Item Number on tender	Precise details of minor deviations included in the offer.

Signature -----
(Manufacturer)

Signature-----
(Vendor / Agent)

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 13-100-3
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**GUARANTEE TABLE FOR
TERMINATION, HEAT SHRINKABLE, ... X .. mm² , 24 KV**

Item	Description
1	Type ----- Catalogue no -----
2	Name of manufacturer -----
3	Max. design voltage ----- kv.
4	Period for which this accessory has been supplied for service conditions similar to Egypt----- years
5	Average splicing time ----- hrs
6	1- Compression lugs, with seal against moisture : - - Material composition (CU%) /(AL%) %/ % - No of compressions ----- - Total length (according to DIN) ----- mm - Plate hole of lug (Dia) ----- mm - Conductor hole inner diam. ----- mm 2- Shrinking tube providing stress control : - - Length/inner diam. as supplied .-----mm/mm - Inner diam / thickness, after free recovery ---- mm/mm 3- Non - tracking shrinking tube : - - Length / inner diam. as supplied ----- mm/mm - Inner diam. / thickness, after free recovery----- mm/mm 4- No of skirts and its diam.----- mm - Length .of leakage path of each skirt ----- mm - With or without adhesive 5- Breakout (spreader cap) : - Dims in -----mm - Total Length / inner dia, for cable side / inner dia, lug side - Inner diam/thickness for cable side after free shrink ----mm/mm - Inner diam/thickness for Lug side after free shrink ----mm/mm 6- Total leakage path of termination----- mm 7- Copper earthing braids ; material composition & weight ---- gm - Total length / cross-sectional area -----mm/mm ² 8- Terminal sleeve (If any) length / inner diam as supplied---- mm/mm
7	Any special tool required -----
8	Any special accessories required
9	Type /Quantity of cleaning solvent -----
10	Length of tin wire solder with flux,----- cm.
11	Weight of silicon grease ----- gm.

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12	Binding tapes : A - Two PVC tapes : - Length / thickness -----m./mm B - Textile tape : - number / length each ----- m.	----- ----- -----
13	Mastic pastes : - A - First type : type / number/ weight. each B - 2 nd type : type / number/ weight. each	----- ----- -----
14	Any other tape : type , length -----	
15	Attach details on rated short circuit current capacity: - a) Copper braid b) Armour c) Bonded braid with armour	
16	Copy of works routine test reports and if requested byEDC also Batch / type test reports on the kits being supplied shall be cleared byEDC prior to shipment confirm compliance -- yes / no	
17	Confirm that visual inspection and routine tests shall be carried out on each kit ----- yes / no	
18	Confirm compliance to the appropriate requirements of Table 2---- yes/no	
19	Marking and packing a) Marking shall be to clause 4.4 -----yes / no b) Packing shall be to clause 4.5 -----yes / no c) Each kit with components shall be complete in itself and separately packed ----- yes / no d) Submit detailed method of packing -----submitted / not submitted.	
20	Guarantee confirm compliance to clause 6.3 ----- yes / no	
21	For direct technical clarification from the manufacturer submit the details mentioned below of the persons responsible : - Name :----- Designation :----- Address :----- Telephone :----- Telex :-----	

we guarantee above mentioned data for our afford material forEDC

vendor / Agent -----

Important : Each tenderer shall submit with his offer the above tender " specifications for cable terminations x mm² 24 KV. With each page signed by the tenderer

This signature means that tenderer confirms verification of his offer for these specification.

Tenders not accompanied with signed specifications shall not be considered.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 13-100-3
CABLE TERMINATION FOR XLPE INSULATED CABLES 12/20 (24) kV (INDOOR/OUTDOOR)	6-05-2025

SCHEDULE GUARANTEE

Schedule of deviations from -----

Technical specification

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Item Number on tender	Precise details of minor deviations included in the offer.

Signature -----

(Manufacturer)

Signature -----

(Vendor / Agent)